

Temporal-Support Identifiability in Event-Detection Evaluation Pipelines

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Abstract

001 An event-detection benchmark identifies each scored sam-
002 ple by one nominal timestamp, but a recurrent detector
003 forms its prediction from a longer history, and the bench-
004 mark does not record how long. We state the missing re-
005 quirement as a property of the protocol, temporal-support
006 identifiability: the benchmark variables must uniquely de-
007 termine the temporal evidence supporting a reported pre-
008 diction. This makes effective temporal support part of the
009 benchmark specification rather than a hidden property of
010 one evaluation implementation. A released SSM-ViT eval-
011 uation pipeline provides a direct counterexample. At the
012 same evaluated timestamp, with frames, weights, labels and
013 evaluator fixed, moving only the chunk boundary changes
014 the recurrent history from one to twenty-one windows within
015 the pooled score. The same move changes S5-B by 4.20
016 mAP over all labeled boxes and by 5.23 ± 0.49 mAP on
017 the velocity-evaluable subset. Because identifiability is a
018 uniqueness property, one controlled pair in which identical
019 recorded protocol variables yield different effective support
020 and different scores is sufficient to show that the protocol
021 under-specifies the score. Two supporting diagnostics re-
022 ported in the supplement, an input-sensitivity measurement
023 on the RVT predictor and an exposure analysis of DSEC
024 label timing, bound the interpretation without being used
025 as recurrent-support estimators. Over all labeled boxes the
026 validation-split ordering of the 5 released checkpoints is it-
027 self reproduced in only 42.5 % of sequence-bootstrap repli-
028 cates, so that ordering cannot certify a timing convention
029 as immaterial. Support-Conditioned AP scores each check-
030 point with the recurrent history fixed to H windows and so
031 equalizes the history across checkpoints. For five released
032 Gen1 checkpoints, S5-B ranks third in the pooled score on
033 the common frames and first at each of $H = 1, 5, 10, 21$,
034 and Kendall's τ between the pooled ordering and the fixed-
035 H orderings lies between 0.00 and 0.40. This is measured
036 for these five checkpoints under this evaluation only. Timing
037 comparisons between recurrent detectors therefore require
038 the support that produced each score to be stated and equal-

ized, a condition an event-detection protocol can record
with its labels.

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1. Introduction

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Two recurrent detectors evaluated at the same nominal
timestamp are not comparable in time unless the history
each one actually used is also identified.

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An event-detection benchmark defines a sample by a
nominal timestamp. A prediction $\hat{s}$ is scored against the
ground truth s^* carried at query time t through a function
 $\rho(\hat{s}, s^*)$ of position, shape and class. Beyond selecting the
label at t , that function contains no variable for the temporal
evidence behind either argument, so any temporal discrep-
ancy enters the score disguised as displacement.

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A recurrent detector does not read only the instant the
benchmark names. It computes $\hat{s}$ from an input window
and from state carried across earlier windows. The label it
is compared against comes from an exposure of finite width,
an annotator or tracker working on the exposed image, and
a geometric transfer into another sensor's frame. The evi-
dence on both sides is an interval, while the protocol records
an instant.

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Three terms carry this throughout. The *nominal times-
tamp* is the instant at which the protocol pairs a prediction
with a label. The *effective temporal support* is the past that
can influence the prediction, including state carried by re-
currence. The *label-acquisition support* is the interval over
which the label's evidence was acquired. A nominal times-
tamp determines neither of the other two uniquely, and a
temporal descriptor may be reconstructable from released
artifacts yet remain unreported or heterogeneous within one
pooled score. We call this protocol property temporal-
support identifiability: whether the released protocol and
artifacts uniquely determine the temporal quantities a re-
ported score represents.

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We test it with a paired intervention on the released
SSM-ViT evaluation pipeline. In that implementation the
nominally carried state is inert across evaluation chunks.

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076	The recurrent history available to a prediction is a deter-	128
077	ministic function of chunk position, and ranges from one	129
078	to twenty-one windows inside one pooled score. Mov-	130
079	ing every chunk boundary 5 windows later re-scores the	131
080	same frames under a longer history at the same evaluated	132
081	timestamp, with frames, weights, labels and evaluator fixed	133
082	(3 574 paired boxes, cluster bootstrap over 406 sequences).	134
083	Effective temporal support alone changes S5-B by 4.20	135
084	mAP over all labeled boxes and 5.23 ± 0.49 mAP on the	
085	velocity-evaluable subset (Sec. 3). A second checkpoint of	
086	the same released pipeline moves by 4.10 and 4.81 points.	
087	Since temporal-support identifiability is a uniqueness prop-	
088	erty, this controlled violation is sufficient to show that the	
089	nominal timestamp does not uniquely specify the tempo-	
090	ral evidence represented by that score. The consequence is	
091	at the level of the protocol rather than of one implementa-	
092	tion: where the protocol does not record the support behind	
093	a score, the reported score is underdetermined. We measure	
094	that condition on the two protocols examined here and do	
095	not assert it of event-detection scores in general.	
096	The counterexample has a corrected counterpart.	
097	Support-Conditioned AP scores a checkpoint with the re-	
098	current history fixed to a stated number of windows H	
099	(Sec. 5). Applied to five released Gen1 checkpoints on	
100	19 946 common frames, it orders them differently from the	
101	pooled score: S5-B is third in the pooled score and first at	
102	each of $H = 1, 5, 10, 21$, and for three pairs the pooled and	
103	$H = 1$ gaps have intervals that exclude zero on opposite	
104	sides. This describes these five checkpoints under this eval-	
105	uation and is not a ranking of the architectures.	
106	Contributions.	
107	• <i>Temporal-support identifiability</i> : an evaluation property	
108	separating the nominal timestamp a protocol records from	
109	the effective temporal support and label-acquisition sup-	
110	port a reported score represents.	
111	• A paired intervention that isolates recurrent support: the	
112	frames, weights, labels, evaluator, and evaluated times-	
113	tamp are fixed, while only the recurrent history changes.	
114	• Its measured consequence: support the protocol does not	
115	record moves a pooled leaderboard-style score by 4.20	
116	mAP overall and 5.23 mAP on the velocity-evaluable	
117	subset. The same benchmark reproduces its validation-	
118	split ordering over all labeled boxes in only 42.5 % of	
119	sequence-bootstrap replicates. Predictor-side and label-	
120	side audits delimit the result.	
121	• A corrected protocol: Support-Conditioned AP, which	
122	fixes the recurrent history behind each score, applied	
123	to five released checkpoints, together with the fields a	
124	benchmark would need to record so that the support be-	
125	hind a score is identifiable.	
126	Work on event-detection timing takes that index as given	
127	and improves what is measured against it. Sensor latency	
	is calibrated away [25, 29], compute latency is folded into	128
	average precision [9, 30, 35], and the temporal window is	129
	redesigned [24, 26]. Each changes what the predictor does	130
	at a fixed benchmark index, whereas the object measured	131
	here is that index itself and the support a score reported un-	132
	der it pools. The two datasets examined here instantiate	133
	this protocol condition. How often it holds elsewhere is not	134
	measured.	135
	Supporting audits. Two audits, summarized in Sec. 4	136
	and reported in Supplement Secs. 15 and 16, delimit this	137
	result and are not used as recurrent-support estimators. Ab-	138
	lating each input bin of the event branch examined here [24]	139
	on real validation samples puts the ablation-sensitivity cen-	140
	troid at -23.81 ms, and the regression-based temporal co-	141
	efficient estimated on nominal-time-matched detections is	142
	-2.40 ± 7.18 ms and is compatible with zero. Both de-	143
	scribe predictor-side timing, and the nominal timestamp en-	144
	codes neither. On the label side, across the eighteen DSEC	145
	training sequences whose exposure files resolve, published	146
	exposure width varies by a factor of 127 against a frame	147
	period fixed at 50 ms, and within-exposure event timing	148
	carries an illumination-locked component, so it is not at-	149
	tributable to object timing without separating that compo-	150
	nent first.	151
	2. Related work	152
	Sensor and runtime latency. Prior work calibrates sen-	153
	sor latency [25, 29], incorporates runtime latency into av-	154
	erage precision [9, 30, 35], or proposes temporal tolerance	155
	[36]. These methods address the time from an event to an	156
	output and take the benchmark’s temporal index as given.	157
	We instead measure that index, testing whether a released	158
	benchmark specification uniquely determines the temporal	159
	quantities behind its score.	160
	Temporal windows and benchmark identifiability.	161
	RVT and SSM-ViT redesign or parameterize temporal	162
	windows [24, 26], while adaptive-window methods com-	163
	pare alternative partitions [28]. Event-RGB frequency	164
	mismatch and exposure duration have also been noted	165
	[22, 23, 37]. Both released implementations leave a	166
	frame’s position inside the evaluation chunk unspecified	167
	by the protocol rather than fixed by the model. The re-	168
	current state entering a chunk is the state saved at the	169
	end of the previous chunk, and is reset only at the first	170
	sample of a sequence. Under streaming sampling ev-	171
	ery timestep emits predictions, while otherwise only the	172
	last does (RVT modules/detection.py:232, 243 and	173
	SSM-ViT :315, 318). LEOD [27] states that event detectors	174
	take only events triggered before the query time and builds	175
	a time-flip augmentation on that premise. It does not mea-	176

177 sure where inside that window the evidence sits. This is a
178 reading of the released code and text, not a measurement
179 of those systems. We read the released per-frame windows
180 and exposures and separate predictor support, output-time
181 interpretation, and label-acquisition support.

182 **Benchmark reliability and evaluation analysis.** Prior
183 work studies detector calibration, stratified model evalua-
184 tion, label-induced benchmark instability, and event-vision
185 evaluation protocols [31–34]. These analyses examine un-
186 certainty or error in reported performance, whereas our fo-
187 cus is the temporal support and acquisition semantics that a
188 nominal detection timestamp leaves implicit.

189 **Label time and attribution.** Waymo and annotation
190 studies estimate per-object capture or annotation offsets [5–
191 7]. Event datasets expose per-box time fields, but their de-
192 grees of freedom are not necessarily specified. LET-3D-AP
193 and TIDE/DETAD/HOTA decompose or bucket spatial er-
194 ror while taking the label clock as given [1, 11–14].

195 **Event benchmarks and exposure.** Gen1, 1 Mpx, DSEC-
196 Det and Ev-3DOD use asynchronous measurements, rec-
197 tified frames, or interpolated labels [16, 19–21]. LEOD
198 establishes causal input windows [27]. The present work
199 reads the released window and exposure artifacts as proto-
200 col evidence and tests whether the resulting temporal quan-
201 tities are uniquely specified.

202 3. Primary evidence: the same-frame chunk- 203 boundary intervention

204 The state the SSM-ViT release carries between evaluation
205 chunks does not affect any of its outputs. It enters as a multi-
206 plicative factor on the first coefficient of an associative scan
207 on which the outputs do not depend. Under the released
208 streaming evaluation this bounds the recurrent history avail-
209 able to a detection to the current 21-window chunk. That
210 history ranges from one window to 21 with the detection’s
211 position in the chunk, and the reported number averages
212 over all 21 positions. Chunk position is a deterministic
213 function of the release’s own index files and is recorded
214 nowhere in the evaluation protocol.

215 **The same-frame intervention.** The intervention moves
216 every chunk boundary 5 windows later and changes nothing
217 else: the same frames, the same weights, the same labels,
218 the same evaluator, and the evaluated timestamp of each
219 frame unchanged. 3 574 labeled boxes are scored twice
220 under this shift, once with one to four windows of recur-
221 rent history and once with seventeen to twenty. They are
222 compared paired within frame with a cluster bootstrap over
223 the 406 sequences. The long-history arm gains 5.23 ± 0.49

mAP points for S5-B and 4.81 ± 0.52 for S5-S on the
velocity-evaluable subset, and over every labeled box the
S5-B gain is 4.20 ± 0.36 points. Mean detection confidence,
which uses no labels, moves with it by 0.0579 ± 0.0048 and
 0.0509 ± 0.0035 . Fig. 1 (bottom) draws one such pair. No
quantity the protocol records distinguishes the two arms.

Graded response and what moves. The size of the
boundary shift is a free parameter. Re-scoring the same
3 574 frames after shifting the boundary so that the starved
block gains +6, +11 and +16 windows of history changes
the velocity-subset mAP of S5-B by 3.73 ± 0.35 , 4.91 ± 0.50
and 5.23 ± 0.49 points, and that of S5-S by 3.54 ± 0.39 ,
 4.29 ± 0.52 and 4.81 ± 0.52 . The zero-shift control, which
re-runs the release’s own boundary placement, returns 0.00.
Confidence on detections matched to a label rises by 0.0409
and 0.0519, and total output falls by 3.49 and 5.09 boxes per
frame, so both terms of precision move. The curve is mea-
sured at four amounts on one dataset and one architecture
family, and a saturation point is not estimated (Supplement
Sec. 13).

The observational contrast across chunk positions. Re-
constructing chunk position over 20 296 labeled frames of
the released evaluation and scoring every labeled box, S5-B
reaches 34.445 mAP on the frames given one to four win-
dows and 40.226 on those given seventeen to 21 (Fig. 1,
top). This contrast compares different frames, so it is dif-
ferenced against the three RVT checkpoints, evaluated on
the same frames with a state that crosses chunk bound-
aries. That controls for frame-position effects shared with
them. It leaves 5.32 ± 0.53 points for S5-B ($z = 10.0$)
and 4.37 ± 0.61 for S5-S ($z = 7.2$), against placebo differ-
ences among the RVT checkpoints of at most 0.67 points
($z = 1.56$). Permuting chunk position within each sequence
puts both SSM values at least 7.6 standard deviations out-
side their own null and every placebo inside it. The two
readings agree in sign and magnitude, and the paired inter-
vention carries the claim because it holds the frames fixed
(Supplement Sec. 13).

The control, measured rather than assumed. That dif-
ferencing treats the RVT checkpoints as a control because
their state survives a chunk boundary. The same-frame in-
tervention applied to RVT measures that property. Driven
on the SSM chunk grid, RVT reproduces its stored per-
window detections bit for bit at zero shift, and the same
3 574 frames are re-scored with every boundary moved 5
windows later. With the released state carried across the
boundary the paired change is -0.01 ± 0.03 points for RVT-
s ($z = -0.3$) and 0.05 ± 0.04 for RVT-b ($z = 1.3$). Drop-
ping the state at every chunk start, the condition the SSM
release is already in, moves the same weights on the same

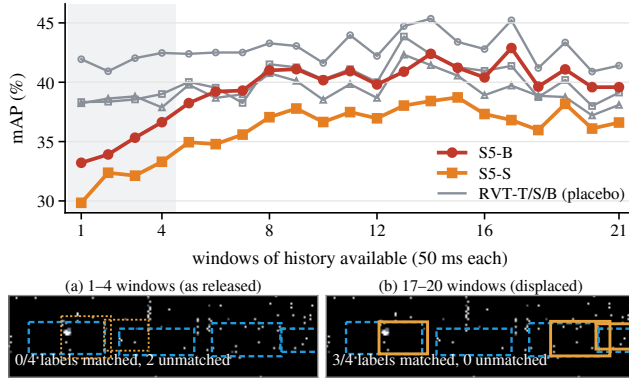


Figure 1. **Top:** benchmark score against the recurrent history the released streaming evaluation made available, at each of the 21 chunk positions, over every labeled Gen1 box with no displacement applied. Shaded: the one-to-four-window block the release scores together with the rest. Between that block and frames given seventeen or more windows, the two SSM checkpoints rise by 5.78 and 4.83 points; the three RVT checkpoints, scored on the same frames with state that crosses chunk boundaries, rise by at most 0.80. **Bottom:** one frame of the same-frame intervention on S5-B. Changed: the chunk boundary, and with it the recurrent history behind the detection. Fixed: frame, weights, labels (dashed), evaluator and evaluated timestamp. Solid: a detection matched to a label of its class at $\text{IoU} \geq 0.5$; dotted: unmatched; drawn above confidence 0.3 and scored at every confidence. The example is illustrative; over all 3574 frames the same intervention changes S5-B by 4.20 mAP over all labeled boxes and 5.23 ± 0.49 mAP on the velocity-evaluable subset (Supplement Sec. 13).

frames by 5.39 ± 0.45 ($z = 12.0$) and 6.91 ± 0.64 points ($z = 10.9$). Random-frame nulls are centered within 0.112 points of zero. The boundary effect follows whether the recurrent state survives the boundary, not the architecture, and the control the difference-in-differences relies on holds as a measurement (Supplement Sec. 13).

What the pooled score reports. One released number therefore pools detections whose recurrent history differs by a factor of 21. Restricting evaluation to the frames receiving 17–21 windows yields an S5-B score 1.75 points above the pooled released-evaluation score. Two systems compared under this protocol at the same nominal timestamps need not have been given the same temporal evidence, and the protocol records no variable by which a reader could tell.

4. Supporting audits

Two audits delimit how the intervention of Sec. 3 is read. Each measures a timing descriptor that the nominal timestamp does not encode, and neither estimates the recurrent support that the intervention manipulates. Their derivations, figures and tables are in Supplement Secs. 15 and 16.

Latent quantities. A benchmark releases one tuple per evaluated instant. The tuple holds a nominal timestamp, the labels asserted at it, the predictions scored against them,

and the score. Three temporal quantities are latent in that tuple and are not recorded in it: the temporal support of the predictor’s input and carried state, the effective state time its output refers to, and the acquisition support of the labels. Call such a quantity *protocol-identifiable* when the released protocol and artifacts determine it uniquely. Under this definition, input and exposure intervals are protocol-identifiable, whereas effective predictor timing, recurrent context represented by a pooled score, and within-exposure attribution are not.

Predictor side. The released RVT configuration [24] indexes each ten-bin, 50 ms input window by its end, and places that end on a label time. The window’s uniform-weight centroid therefore lies 25 ms before the instant at which the label asserts a state. Ablating each bin of the released `rvt-t` checkpoint over 480 Gen1 validation samples, with the incoming recurrent state held fixed, puts the ablation-sensitivity centroid at -23.81 ms. A regression of detection-center residuals on label velocity, which carries the temporal calibration of [8] to a box center and is fitted on nominal-time-matched detections, gives a coefficient of -2.40 ± 7.18 ms. The two are distinct operational summaries. Their point estimates are 26.2 ms apart, and the difference is not an estimate of architectural latency. Over a 1000 ms horizon the newest window accounts for 22.7 % of the summed window-ablation sensitivity, so the newest-window centroid does not describe the full recurrent support. Displacing ground-truth centers across the measured label-to-centroid interval changes mAP on the velocity-evaluable subset by 0.04 points and does not reorder the 5 released checkpoints, an effect at the benchmark’s sampling-resolution floor. That near invariance follows from Gen1’s speed composition. It characterizes the metric’s resolution and does not show the absence of a temporal discrepancy (Supplement Secs. 2, 11–12, 14, 15).

Label side. Across the eighteen DSEC training sequences whose exposure files resolve [15], published exposure width varies by a factor of 127 against a frame period fixed at 50 ms, and 32.5 % of frames in six sequences are pinned at an auto-exposure ceiling of $14\,996\,\mu\text{s}$. In those sequences the event-rate spectrum carries the 100-Hz line and its harmonics, which the daytime control lacks. Within-box dispersion of event time inside one exposure is $206.6\,\mu\text{s}$ against an analytic null of $93.0\,\mu\text{s}$. A frame-level permutation null gives $78.6\,\mu\text{s}$, and the excess remains after the clean-pixel control. Random-position boxes show that the dispersion is not specific to annotated object regions. Within-exposure event timing therefore contains an illumination-locked component, and it cannot be attributed to object timing without separating that component first (Supplement Secs. 9 and 16). DSEC-Det ships more than

one annotation version [18], and Supplement Sec. 8 names the one measured here.

5. Support-conditioned AP

Sec. 3 shows that one released protocol pools detections that received different histories. This section scores the 5 released Gen1 checkpoints at a stated history and compares the result with the pooled score. It follows the counterexample with the corresponding protocol.

Protocol. For a labeled frame scored at window r and a history length H , the detector starts from a reset recurrent state at window $r - H + 1$ and runs through window r , and the detections of window r are scored with the labels, confidence threshold, NMS and evaluator of the released-protocol dumps. We call the resulting AP at fixed H Support-Conditioned AP. H is the number of recurrent windows allowed before scoring, H_{nominal} . A frame with fewer than H earlier windows receives $H_{\text{forced}} = \min(H, r + 1)$ windows instead. All checkpoints are scored on the same 19 946 frames, those with $H_{\text{forced}} = H_{\text{nominal}}$ at every $H \in \{1, 5, 10, 21\}$. The other 350 of the 20 296 labeled frames have fewer than 21 windows of history and are excluded. The common frames lie in 406 sequences. The column *pooled* is the released evaluation of each checkpoint, in which the history depends on the release and on the frame, restricted to the same common frames. The score of the released evaluation on all 20 296 frames is listed separately. The access log of every run shows that the number of distinct windows read equals H_{forced} and that the first window read is $r - H_{\text{forced}} + 1$ in every cell of every checkpoint (Supplement Sec. 13).

Result. Table 1 lists the scores. On the common frames the pooled ordering is RVT-b, RVT-s, S5-B, RVT-t, S5-S. S5-B is first at each of $H = 1, 5, 10, 21$. At $H = 1$ the ordering is S5-B, S5-S, RVT-b, RVT-s, RVT-t, and at $H = 21$ it is S5-B, RVT-b, RVT-s, S5-S, RVT-t. Kendall’s τ between the pooled ordering and the ordering at $H = 1, 5, 10, 21$ is 0.00, 0.20, 0.20 and 0.40. The pooled ordering is reproduced in 48.5 % of the bootstrap draws, whereas the rate for all labeled boxes in Supplement Sec. 11 is 42.5 %. The fixed- H orderings are reproduced in 79.5, 87.0, 72.5 and 43.5 % of draws.

Several pairwise gaps change sign. RVT-t minus S5-S is +3.40 [+2.58, +4.21] points in the pooled score and -2.64 [-3.64, -1.96] at $H = 1$. RVT-s minus S5-S is +4.20 [+3.35, +4.88] pooled and -1.76 [-2.66, -0.95] at $H = 1$. RVT-b minus S5-B is +3.44 [+2.54, +4.20] pooled and negative at every H , down to -0.75 [-1.66, -0.12] at $H = 21$. For RVT-s and RVT-t against S5-B the pooled intervals include zero (+0.78 [-0.15, +1.83] and -0.02

[-0.98, +1.14]), while at every H the point gaps lie between -4.96 and -2.98 and between -5.84 and -3.52 and the intervals exclude zero. The pairwise gaps for all pairs and the curves over H are in Supplement Sec. 13. For the RVT checkpoints the pooled score exceeds the $H = 21$ score (RVT-b: 42.36 against 39.24), and for the S5 checkpoints it is below it (S5-B: 38.92 against 40.00). The RVT release carries its state beyond 21 windows, which the fixed- H protocol does not.

What the comparison supports. H is the history allowed before scoring, not a measurement of how many windows influenced the detections. The access log shows which windows were read. In an oldest-window probe, replacing the oldest admitted window changed the output in 1217 of 1218 probed cells for four checkpoints and in 1216 of 1218 for RVT-t. Every H starts the RVT checkpoints from a reset state, which the RVT release does not do at evaluation: it carries its state and clears it only at the first sample of a sequence. A short H therefore combines a shorter history with a reset the checkpoint did not see in this form, and the experiment does not separate the two. Supplement Sec. 13 reports that RVT-s and RVT-b score 5.39 and 6.91 points higher on the velocity-evaluable subset when a reset block is given seventeen to twenty windows instead of one to four. The SSM-ViT release already evaluates each chunk from a reset state (Sec. 3). The ordering change is therefore a result for these five checkpoints under this evaluation. It does not rank the architectures, and it does not show that either family is better in general.

What a benchmark should record. The three quantities of Sec. 4 are not recoverable from a nominal timestamp, and the intervention above shows a case in which the missing one changes the score. We propose that a benchmark record, for each scored sample, five fields. The *nominal timestamp* is the instant at which the sample is scored. The *input window* is what the detector reads at that timestamp. The *state-reset policy* states where the recurrent state is cleared, including chunk boundaries and sequence starts. The *effective history length* is the number of windows read, as the access log of the evaluation run records it. The *label-acquisition support* is the interval over which the label’s evidence was acquired. Together, the reset policy and the history length determine the effective temporal support. A score is then reported at a declared history length, as in Table 1, together with the distribution of history lengths that the pooled score contains. These fields do not equalize the support across checkpoints. They record it, so that scores can be compared at a history that both detectors received.

	pooled	$H = 1$	$H = 5$	$H = 10$	$H = 21$	pooled, all frames
RVT-t	38.91 [36.94, 41.13]	27.85 [25.70, 29.99]	33.36 [31.19, 35.51]	35.15 [33.01, 37.52]	36.47 [34.34, 38.91]	38.90
RVT-s	39.70 [37.40, 42.07]	28.72 [26.57, 30.95]	34.06 [31.69, 36.23]	35.75 [33.23, 37.98]	37.02 [34.61, 39.36]	39.68
RVT-b	42.36 [39.85, 44.28]	30.25 [27.91, 32.29]	35.99 [33.78, 37.93]	37.86 [35.47, 39.86]	39.24 [36.72, 41.23]	42.32
S5-S	35.50 [33.44, 37.78]	30.48 [28.73, 32.72]	34.57 [32.65, 36.84]	35.97 [33.92, 38.17]	36.95 [34.89, 39.18]	35.55
S5-B	38.92 [36.70, 40.84]	33.69 [32.00, 35.59]	38.07 [36.03, 40.02]	39.45 [37.15, 41.42]	40.00 [37.78, 41.97]	38.95
Kendall τ against pooled	1.00	0.00	0.20	0.20	0.40	
ordering reproduced (%)	48.5	79.5	87.0	72.5	43.5	

Table 1. mAP (points) of the five released Gen1 checkpoints on the 19946 common frames, over every labeled box, with the recurrent history fixed to H windows (reset state, exactly H windows read) or as given by the released evaluation (pooled). Brackets: 95 % interval from 200 cluster-bootstrap draws over the 406 sequences, the same draw for every checkpoint and column. Last column: the released score on all 20 296 frames. The final rows give Kendall’s τ between the ordering of each column and the pooled ordering, and the share of bootstrap draws whose full ordering equals that column’s point ordering.

6. Scope and limitations

Each supporting audit (Sec. 4, Supplement Secs. 15 and 16) is conditional on its stated instrument: local window-ablation sensitivity, nominal-time association under the reported regression specification, and a local constant-velocity displacement surrogate. The primary chunk-boundary intervention is established on two released SSM-ViT checkpoints under their released streaming evaluator, the second testing that the violation is not confined to one weight set. Gen1 and DSEC are separate timing audits, not premises of that counterexample.

Support-Conditioned AP (Sec. 5) covers five released Gen1 checkpoints on the 19 946 frames that have at least 21 windows of history, at $H \leq 21$, with a reset state at every H and intervals from 200 sequence-bootstrap draws. It does not test $H > 21$, and it does not measure how many windows influenced a detection. The orderings it reports hold for these checkpoints under this evaluation, and no claim about the two architecture families in general follows from them.

7. Conclusion

At the same evaluated timestamp, with frames, weights, labels and evaluator fixed, moving only the chunk boundary shifts S5-B by 4.20 mAP over all labeled validation boxes and 5.23 mAP on the velocity-evaluable subset. One pooled score in the released SSM-ViT pipeline combines predictions carrying one to twenty-one windows of recurrent history. The chunk boundary is an unreported degree of freedom of the protocol. It changes the recurrent support behind the reported score while every recorded quantity stays fixed, which is exactly a failure of temporal-support identifiability.

The result does not single out a canonical temporal-support statistic, nor attribute the shift to one stage of the pipeline. The predictor-side and label-side audits stay descriptors of timing rather than estimates of recurrent support.

Support-Conditioned AP is the corrected counterpart of the counterexample: it fixes the recurrent history behind each score. For the five released Gen1 checkpoints on

the 19946 common frames, S5-B ranks third in the pooled score and first at each of $H = 1, 5, 10, 21$, the Kendall τ between the pooled and fixed- H orderings lies between 0.00 and 0.40, and for three pairs the pooled and $H = 1$ gaps have intervals that exclude zero on opposite sides. Part of the short- H deficit of the RVT checkpoints may come from the reset that every H imposes on them, and the experiment does not separate it from the shorter history. These statements hold for these checkpoints under this evaluation.

Where a protocol does not record the support behind a score, the reported score is underdetermined. Two scores at the same nominal timestamp are not performance at the same effective time when the recurrent support behind them differs. Comparisons across recurrent architectures should therefore report or control the three latent temporal quantities alongside the score. Concretely, a benchmark can record the nominal timestamp, the input window, the state-reset policy, the effective history length and the label-acquisition support of each scored sample, and report the score at a declared history length.

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